We work with the **Yang–Nevzorov model with full observations**:

$$\ell(\gamma,\mu,\sigma)=\sum_{t=1}^{T}\left[t\log\gamma+\log f(x_t)+(\gamma^t-1)\log F(x_t)\right]$$

✅ 1. Gumbel distribution

$$f(x)=\frac{1}{\sigma}\exp\left(-\frac{x-\mu}{\sigma}\right)\exp\left(-e^{-\frac{x-\mu}{\sigma}}\right)$$

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Define:

$$u_t=\frac{x_t-\mu}{\sigma},\quad e_t=e^{-u_t}$$

✅ 2. Log-likelihood (simplified)

Substitute:

$$\log f(x_t)=-\log\sigma-u_t-e_t,\quad \log F(x_t)=-e_t$$

Final expression

$$\boxed{\ell(\gamma,\mu,\sigma)=\sum_{t=1}^{T}t\log\gamma-T\log\sigma-\sum_{t=1}^{T}u_t-\sum_{t=1}^{T}\gamma^t\,e_t}$$

where

$$u_t = \frac{x_t - \mu}{\sigma}, \qquad e_t = \exp(-u_t),$$

define

$$w_t = \gamma^t.$$

Then

$$\ell = S_t \log \gamma - T \log \sigma - \sum u_t - \sum w_t e_t,$$

with

$$S_t = \sum_{t=1}^{T} t = \frac{T(T+1)}{2}.$$

✅ 3. First-order derivatives

- **w.r.t. μ**

$$\boxed{\ell_\mu = \frac{T}{\sigma} - \frac{1}{\sigma}\sum \gamma^t e_t}$$

- **w.r.t. σ**

$$\boxed{\ell_\sigma = -\frac{T}{\sigma} + \frac{1}{\sigma^2}\sum (x_t - \mu) - \frac{1}{\sigma^2}\sum \gamma^t (x_t - \mu) e_t}$$

- **w.r.t. γ**

$$\boxed{\ell_\gamma = \sum_{t=1}^{T} \frac{t}{\gamma} - \sum_{t=1}^{T} t\,\gamma^{t-1} e_t}$$

✅ 4. Second-order derivatives

- $\mu\mu$

$$\ell_{\mu\mu} = \quad -\frac{1}{\sigma^2}\sum \gamma^t e_t$$

- $\sigma\sigma$

$$\begin{aligned}\ell_{\sigma\sigma} = \frac{T}{\sigma^2} - \frac{2}{\sigma^3}\sum(x_t - \mu) \\ + \frac{2}{\sigma^3}\sum \gamma^t (x_t - \mu) e_t \\ + \frac{1}{\sigma^4}\sum \gamma^t (x_t - \mu)^2 e_t\end{aligned}$$

$$\frac{\partial^2 \ell}{\partial \sigma^2} = \frac{T - 2\sum u_t + \sum w_t u_t (2 - u_t) e_t}{\sigma^2}$$

- $\gamma\gamma$

$$\ell_{\gamma\gamma} = -\sum \frac{t}{\gamma^2} - \sum t(t-1)\gamma^{t-2} e_t$$

- **Cross derivatives**

$\gamma\mu$

$$\boxed{\ell_{\gamma\mu} = \frac{-1}{\sigma} \sum t\,\gamma^{t-1} e_t}$$

$\gamma\sigma$

$$\boxed{\ell_{\gamma\sigma} = \frac{-1}{\sigma^2} \sum t\,\gamma^{t-1} (x_t - \mu) e_t}$$

$\mu\sigma$

$$\boxed{\ell_{\mu\sigma} = -\frac{T}{\sigma^2} + \frac{1}{\sigma^2} \sum \gamma^t e_t - \frac{1}{\sigma^3} \sum \gamma^t (x_t - \mu) e_t}$$

$$\boxed{\frac{\partial^2 \ell}{\partial\mu\partial\sigma} = \frac{-T + \sum w_t e_t (1 - u_t)}{\sigma^2}}$$